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A Perusal of Machine-Learning Algorithms in Crop-Yield Predictions

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6.1 INTRODUCTION

Agriculture is crucial in resolving urgent societal issues, particularly in the context of food provision. Currently, a significant section of the world population suffers from hunger, which is mostly caused by food shortages and insufficiencies, which are aggravated by a continually expanding population [1]. The consequences of population increase, natural temperature changes, soil erosion, and changing climate-related needs necessitate novel techniques to ensure constant and timely food cultivation and output. Furthermore, there is an urgent need to promote sustainability in agricultural food production [2]. These requirements highlight the critical need for land evaluation, crop protection, and accurate crop-yield projection in increasing global food production [3]. Crop productivity is closely connected to a slew of factors, including topography, soil quality, insect pressures, genetic characteristics, water availability, weather patterns, and harvest planning [4]. Crop-yield estimation procedures and approaches are both time-specific and fundamentally nonlinear. The integration of a broad variety of interdependent aspects, many of which are controlled by uncontrollable external forces, adds to the complexity [5].

Farmers have traditionally relied on personal experiences and historical data to anticipate crop yields and make important production decisions based on these forecasts. However, in recent years, novel technologies such as crop model simulations and machine learning (ML) have emerged, with the potential for more accurate yield estimates and the capacity to analyze

enormous datasets utilizing high-performance computers. Several research studies are now revealing the greater potential of ML algorithms in this domain as compared to traditional statistical techniques [6].

A subset of artificial intelligence (AI), ML, allows computers to learn without explicit programming. These strategies overcome the limits of both linear and nonlinear agricultural systems, improving predictability. ML models learn by being exposed to certain tasks and patterns in training data. Following that, these models apply their learned information to fresh data, assuming that the knowledge obtained during the training phase is transferrable and relevant. A profusion of instructive and unique research, as well as reviews, have been published in the field of agricultural output assessment over the last 15 years [7] and conducted a thorough assessment of practical remote sensing applications in palm oil farming, finding research gaps and providing viable solutions. It should be noted, however, that their study did not primarily focus on predicting techniques for palm oil production. Researchers [8] investigated contemporary developments in official statistics generation, including remote sensing, surveys, and the integration of multiple data sources, such as meteorological and administrative data. While their research identified the potential to improve current crop production forecasting methods, it also acknowledged the inherent uncertainty involved with such forecasts. Nonetheless, their article, although examining agricultural yield strategies [9], provided a well-structured review that delves into a variety of features and prediction techniques. The focus of their essay, however, was on information extraction rather than identifying research gaps, making recommendations, or critically analyzing previous work. Their research advised a thorough investigation of new parameters for forecasting agricultural production. Another study [10] looked at palm oil productivity based on physiological plant variables, with the goal of providing insights into the causes causing palm oil output gaps. The researchers [11] concentrated on the practical use of ML in agriculture. Within this framework, they examined studies on animal, crop, water, and land management. Another study [12] attempted to determine the ripeness of fruits in order to optimize harvest scheduling and production forecasts.

ML is a subset of AI that focuses on developing efficient approaches for yield prediction based on a wide range of influencing factors. ML extracts important insights from datasets by identifying observable patterns and relationships within the data [13]. These ML systems require training with

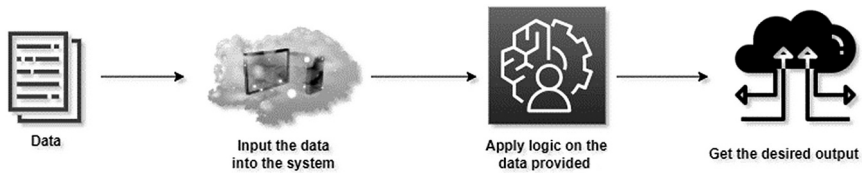


FIGURE 6.1

An ML technique.

datasets that contain outcomes resulting from previous experiences in order to develop prediction models.

The construction of the predictive system requires the use of several characteristics, and the system's parameters are defined using historical data during the training phase [14]. Following that, in the testing phase, a subset of the past data that was not used during training is used to evaluate the system's development. Figure 6.1 depicts an ML technique.

6.2 LITERATURE REVIEW

6.2.1 Crop-Yield Prediction Methods: Traditional vs. Machine Learning

In the past, crop-yield prediction relied more on manual approaches rather than technology-driven methods. These traditional techniques encompassed practices like referencing historical data, monitoring weather conditions, conducting soil analysis, and performing field inspections, among others. However, a notable drawback of these conventional methods was their sluggishness, limited accuracy, and the time they consumed. Numerous sectors employ ML techniques, including supermarkets using ML to forecast customer phone usage and analyze consumer behavior. In the realm of agriculture, ML has been a long-standing tool. One of the complex challenges in precision agriculture revolves around predicting crop production, and various models have been developed and demonstrated their effectiveness in addressing this challenge. Given the vast array of factors influencing agricultural production, such as climate, weather, soil conditions [15], fertilizer application, and seed varieties, tackling this issue necessitates the utilization of multiple datasets. This underscores that the prediction of agricultural yields entails a multifaceted and intricate process, far from being a straightforward task [16]. Present-day

crop-yield prediction models exhibit a reasonably accurate ability to anticipate actual yields, although there remains a continuous pursuit for higher prediction performance [17]. We undertook a comprehensive analysis of the existing literature through a systematic literature review (SLR) to gain a comprehensive understanding of the research conducted on the utilization of ML [18] in predicting agricultural production. Such a rigorous SLR study [19] not only brings forth fresh perspectives but also enhances the comprehension of the current state of the art, which is particularly beneficial for newcomers entering the field of research.

6.2.2 Key Challenges in Crop-Yield Prediction

While ML holds significant potential for enhancing crop analysis and prediction, it is essential to account for various challenges that must be addressed.

- i. *Data Volume*: A significant amount of data is frequently required for ML models to be trained effectively. Large dataset management and collection might be challenging in the agriculture industry, especially for small farms [20].
- ii. *Data Quality*: The caliber of the training data determines how accurate and reliable ML models are. Due to changes in the soil, climate, geography, and other environmental factors, collecting high-quality data in agriculture can be difficult. As a result, gathering and processing data might be labor-intensive. The main difficulties with fruit identification and recognition using deep learning are explored in reference [21]. They found that several issues, including the dearth of high-quality fruit datasets, the detection of small target fruits, fruit identification in obscured and densely populated environments, the detection of fruits with varying sizes and species, and the development of lightweight fruit detection models are to blame for low accuracy, slow processing speed, and limited robustness in fruit detection and recognition.
- iii. *Model Complexity*: Given the complexity of agricultural systems, constructing ML models that comprehensively consider all of the significant variables influencing the development and yield of crops can be a daunting task. Among the most often used approaches are neural networks (NN), random forests (RF), support vector machines (SVM), decision trees (DT), and Naive Bayes algorithms.

These algorithms have considerable hurdles, most notably dealing with large datasets, which increase training time complexity and processing needs, especially in the case of SVM. Furthermore, modifying the algorithm for each situation is difficult in the setting of RF [22]. The creation of a solid big data architecture is one of the most difficult challenges, needing both flexibility and great scalability; researchers [23] examined the parameters controlling soil temperature and determined that the link between factors influencing soil temperature is complex and demanding.

- iv. *Accessibility*: Securing the required hardware and software infrastructure for the development and deployment of ML models can present difficulties when resources are constrained.
- v. *Human Factor*: It may take additional time for farmers and other stakeholders to prepare themselves for adopting novel techniques and technologies, such as ML-based systems. To encourage wider adoption, technology should be designed to be more user-friendly, accessible, and capable of delivering tangible benefits.

6.3 COLLECTION AND PREPROCESSING OF DATA

6.3.1 Data Sources: Remote Sensing, Internet of Things, Weather Stations, and So On

The systematic gathering and measuring of information relating to variables' interests are what data collection includes. This procedure allows researchers to address specific study questions, test hypotheses, and evaluate outcomes. The act of gathering data is a key part of research that cuts across academic fields such as natural and social sciences, humanities, and business. Although the individual methods used vary depending on the field, the overriding aim of guaranteeing precise and accurate data collection remains constant.

i. *Data Collection Using Remote Sensing*

Evaluating the well-being of crops throughout their growth stages: making predictions about their yield. Remote sensing plays a role in monitoring the health of crops in a specific field on an ongoing basis. Its main purpose is to detect any threats and take measures before

they cause any harm. Additionally [11], it serves as a method for estimating the yield of the cultivation cycle and determining the time for harvesting based on weather conditions. CropIn's SmartRisk is an example of a product that utilizes satellite-based sensing technology combined with processing capabilities to provide accurate predictive insights about the entire cultivated area. Remote sensing has the enormous advantage of allowing growers to continually monitor their fields independent of their physical location. This is especially beneficial for large agribusinesses with farms scattered across the country since all farms must produce products of consistent quality [24]. Remote sensing allows all stakeholders to work together to follow the crop's progress through various phases, ensuring that it satisfies the business's requirements and maintains uniform quality. Crop management software, such as CropIn's SmartFarm, uses remote sensing technologies to give detailed insights about crop development and performance throughout the growing season. Researchers [25] provide further information about remote sensing and its application in agriculture.

ii. *Data Collection Using the Internet of Things*

Collecting Internet of Things (IoT) data involves the utilization of sensors to oversee the operation of IoT-connected devices. These sensors continuously observe the status of the IoT network, gathering and transmitting real-time data that can be stored and retrieved whenever necessary. There are different types of IoT data. Different types of IoT data are mentioned next:

- *Submeter Data:* Property owners can utilize submetering to measure individual utility use in multi-user environments. Submeters can collect data in buildings with several tenants using water, electricity, gas, or cable resources.
- *Environmental Data:* IoT sensors can measure and monitor environmental data such as humidity, temperature, movement, and air quality.
- *Equipment Data:* It includes the data that pertains to the state of IoT devices. Equipment data is collected in real time to allow predictive maintenance tasks.
- Current uses provide insights into the effects of the IoT and new practices. Nonetheless, with the advancement of technology [26], IoT technologies may play a critical part in numerous farming tasks. These include using communication infrastructure, data

acquisition, smart objects and sensors, mobile devices, exploiting cloud-based intelligent information, easing decision-making, and automating agricultural activities. IoT technology oversees the remote monitoring of plants and animals while collecting data from mobile phones and gadgets [27]. Researchers have put forth diverse approaches, architectural solutions, and a range of equipment to monitor and transmit crop-related information across different growth stages tailored to various crop types and field conditions. Numerous manufacturers offer communication devices, multiple sensors, robotic systems, heavy machinery, and drones designed for data collection and dissemination purposes. Additionally, food and agriculture organizations, in collaboration with government bodies, are actively formulating guidelines and policies to regulate the use of these technologies, intending to ensure food safety and environmental preservation [28].

iii. *Data Collection Using the Weather Stations*

Weather stations are buildings or objects with a variety of equipment and sensors that gather and measure meteorological information and local environmental variables. These stations are positioned carefully to keep track of variables, including temperature, humidity, atmospheric pressure, wind direction, speed, precipitation (rain and snow-fall), and, occasionally, even more parameters like solar radiation and soil moisture. Simple home weather stations for personal use can range in complexity from highly sophisticated automated systems employed by meteorological authorities and research institutions. Weather data is frequently made publicly available for use in a variety of applications and decision-making processes by modern weather stations that frequently send data in real-time to centralized databases.

Farmers need to learn how to monitor environmental conditions properly and effectively, especially for plants that are particularly sensitive to the weather, to reduce the effects of climate change on diverse crops. Two popular techniques for gathering information and monitoring weather conditions are on-site sensors and weather stations. Although sensors can gather precise meteorological data locally, installing and maintaining them may be expensive and time-consuming. Using online weather stations, which are often controlled by the government and accessible to everyone, is an option; nevertheless, their accuracy is debatable because of how frequently they are situated far [29] from farmers' greenhouses. As a result, we contrasted

the precision of kriging estimators utilizing data from local sensors and weather station data gathered by the Central Weather Bureau. Temperature data were interpolated using the spatiotemporal kriging technique. For comparison, the actual value in the greenhouse's center was utilized. Our findings showed that the local sensor estimator had somewhat higher accuracy than the weather station estimator. On-site sensors can provide farmers with precise estimates of environmental data, but if they are not accessible, utilizing a nearby weather station estimation is also acceptable (Figure 6.2) [30].

6.3.2 Cleaning, Transformation, and Integration of Data

Preparing raw data for analysis and model training by cleaning and pre-processing it are crucial phases in the ML pipeline. These stages are essential since your data's applicability and quality have a big impact on the effectiveness and precision of your ML models.

- i. **Data Cleaning:** Finding and fixing flaws or inconsistencies in your dataset is known as data cleaning. It strives to guarantee the accuracy and dependability of your data. Data cleansing activities frequently involve the following:

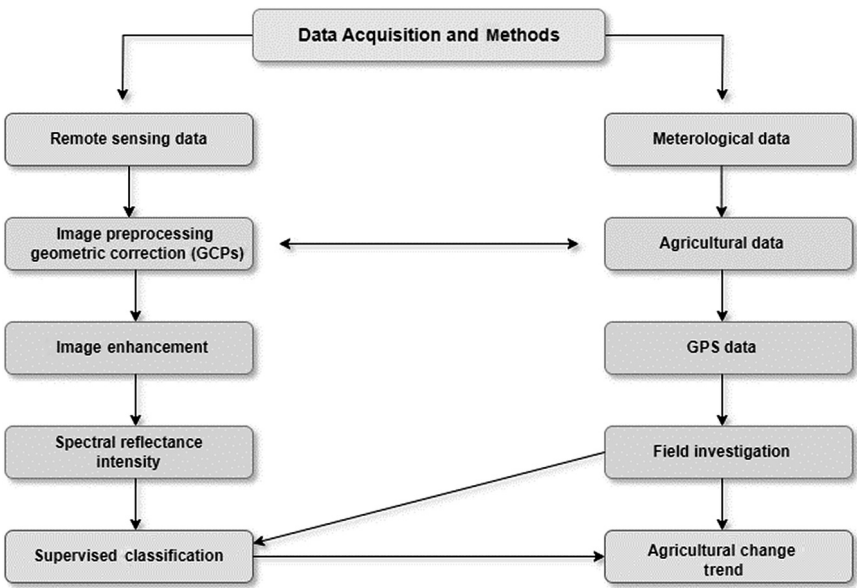


FIGURE 6.2
Various methods of data collection.

- *Handling Missing Values*: Removing rows with missing values, imputing missing values with a default value or a statistical measure (such as the mean or median), or using more sophisticated imputation techniques are all examples of how to handle missing values.
 - *Outlier Detection and Treatment*: Determining whether to eliminate, transform, or leave as-is outliers (data points notably different from the majority) based on the context.
 - *Data Formatting*: Data formatting involves making sure that the data types are acceptable and consistent, converting them if required, and formatting the data according to the needs of the ML algorithm.
 - *Handling Duplicates*: Detecting and getting rid of duplicate entries will prevent bias and redundancy in the dataset.
 - *Noise Reduction*: Data smoothing or filtering techniques are used to reduce noise, especially in time-series or sensor data. Researchers [31] use big data with random sample blocks to explain the process of data cleaning and preprocessing.
- ii. **Data Preprocessing**: The translation of cleansed data into a format appropriate for ML algorithms is known as data preparation. Researchers [32] explain the process of data preprocessing combinations to improve the performance of quality classification in the manufacturing process. Several tasks are included in this process, including the following:
- *Feature Scaling*: Feature scaling is the process of reducing numerical characteristics to a common range (e.g., normalization or standardization) to prevent features from dominating others during model training.
 - *Feature Engineering*: This is the process of developing new features or modifying existing ones to extract useful information or enhance model performance. This includes the one-time encoding of categorical variables, the creation of interaction features, and the extraction of date/time features.
 - *Data Encoding*: Transforming variables that are categorical into numerical format using techniques such as one-hot encoding or label encoding.
 - *Normalization*: Normalization is the process of scaling features to have a zero mean and unit variance so that ML algorithms can converge faster [33] (Figure 6.3).

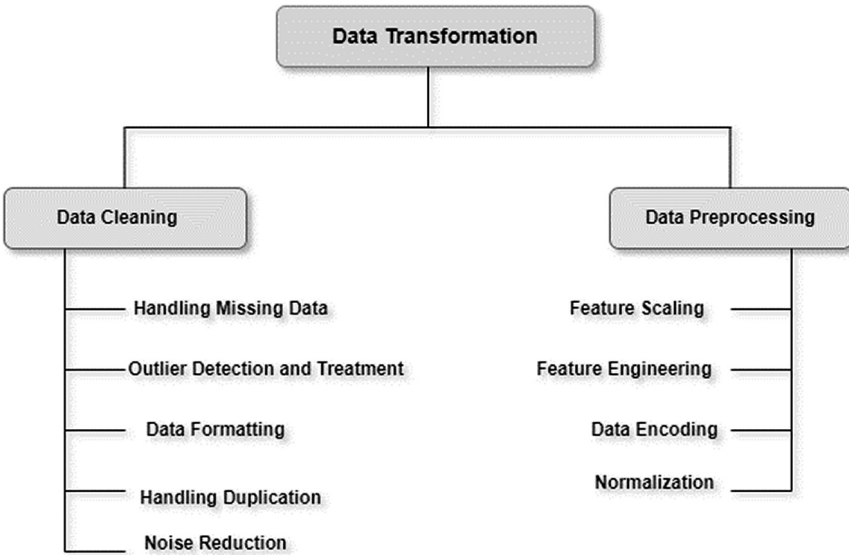


FIGURE 6.3
Methods of data cleaning and preprocessing.

6.3.3 Crop-Yield Prediction Using Feature Engineering

Feature design or engineering is an important stage in preparing data for ML algorithms. It entails generating new features (also known as variables or attributes) from current data to improve the performance of an ML model [34]. The purpose of feature engineering is to extract useful information from raw data and represent it in a manner that the ML algorithm can learn from it successfully. Feature engineering helps us to extract the dependent and independent variable that fits the ML model and helps us in the further fine-tuning of preconceived beliefs and probability. Researchers [35] use the weather data and the historical data of maize and the Irish potato for future prediction of the yield of the Irish potato and maize (Figure 6.4).

6.4 MACHINE-LEARNING ALGORITHMS FOR CROP-YIELD PREDICTION

ML algorithms are software programs that can detect hidden patterns in data, make predictions, and improve their performance by self-learning

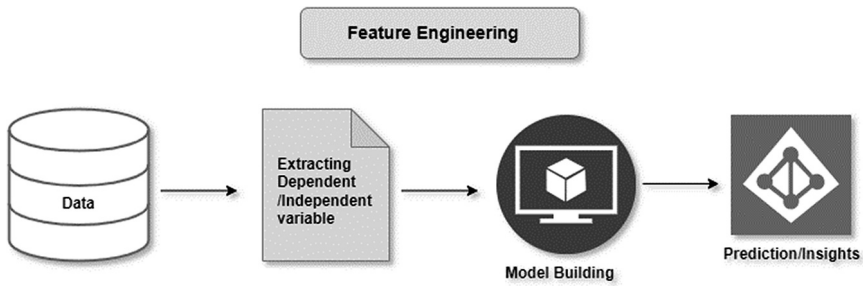


FIGURE 6.4

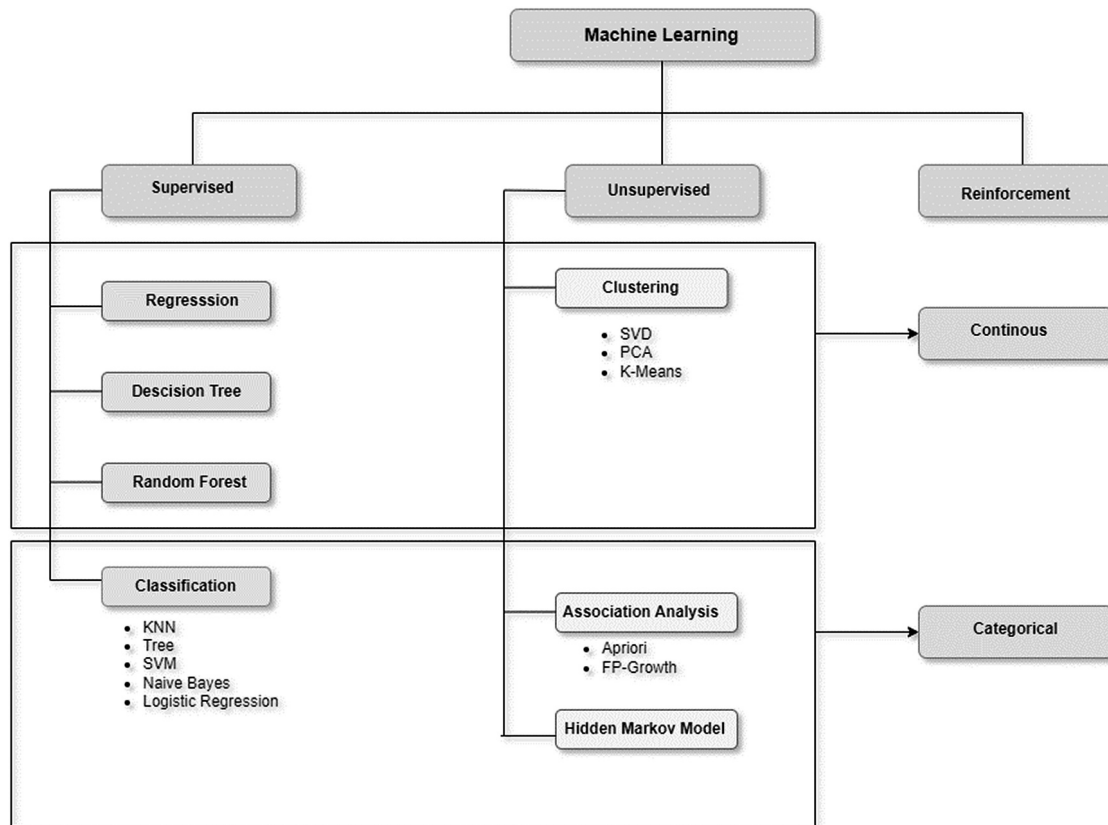
Feature engineering.

from experience. ML employs a variety of algorithms, each adapted to a certain purpose [36]. Simple linear regression, for example, is well-suited for prediction tasks such as anticipating stock market movements, but the K-nearest neighbors (KNN) method is well-suited for classification challenges. The ML tasks are majorly divided into classification and regression tasks.

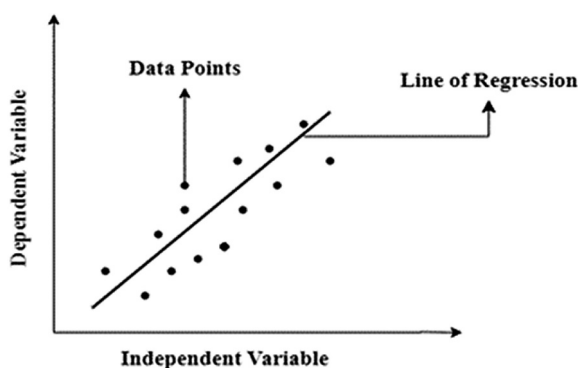
- a. **Classification Task:** Classification tasks are the tasks that necessitate the application of ML algorithms to learn how to assign a class label to problem domain instances [37]. One simple example is categorizing emails as “spam” or “not spam.” The most popular classification algorithms are logistic regression, Naive Bayes, KNN, and SVM.
- b. **Regression Task:** Regression represents a supervised ML approach for forecasting continuous values. The regression algorithm’s ultimate purpose is to plot the best-fit line or curve between the data. Variance, bias, and error are the three major metrics used to evaluate the trained regression model [38]. The most popular regression algorithms are Linear regression, Ridge regression, Lasso regression, Polynomial regression, and Bayesian linear regression. More broadly, ML is divided into the following categories: Supervised learning, Unsupervised learning algorithms, and Reinforcement learning algorithms (Figure 6.5).

6.4.1 Linear Regression Models

The linear regression technique is one of the most basic and widely used ML methods. This is a statistical approach for performing predictive analysis. Linear regression forecasts continuous/real or quantitative variables

**FIGURE 6.5**

Different types of ML algorithms.

**FIGURE 6.6**

Linear regression

such as sales, salary, age, product price, and so on. The linear regression technique demonstrates a linear connection between a dependent ($\hat{y}$) variable and one or more independent (X_1) variables, thus the name. Because linear regression demonstrates a linear connection, it determines how the value of the dependent variable evolves concerning the value of the independent variable [38]. The linear regression model represents the association between variables using a slanted straight line. Mathematically linear regression can be represented as: $\hat{y} = b_0 + b_1 X_1$, where $\hat{y}$ = Dependent Variable; b_0 = Y – Intercept; b_1 = Slope Coefficient; X_1 = Independent Variable (Figure 6.6).

6.4.2 Decision Tree and Random Forest Models

Decision Tree: A DT is a method of supervised learning that may be used to solve classification and regression problems, while it is mostly used to solve classification problems. It works as a treelike classifier, with core nodes representing dataset attributes, branches representing decision rules, and each leaf node representing a result (Rokach & Maimon, n.d.). There are two types of nodes in a DT: decision nodes and leaf nodes. Decision nodes are used to make decisions and have several branches, whereas leaf nodes indicate the results of those decisions and do not branch out any further. Inside a DT, the algorithm initiates its process at the root node, progressing toward predicting the class for a provided dataset. This procedure involves examining the root node's property values against the attributes of the actual dataset record, determining the appropriate branch to follow, and proceeding to the next node based on the outcome of this comparison.

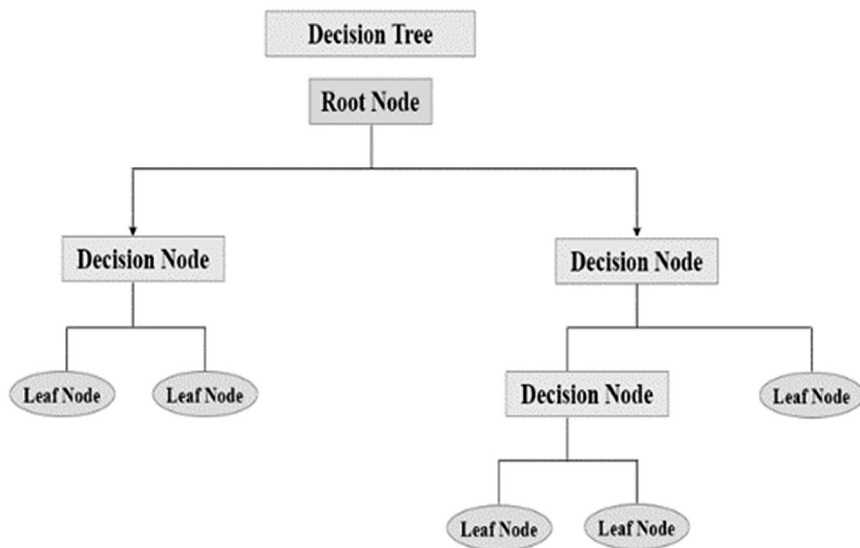


FIGURE 6.7
Decision tree.

The CART algorithm, which stands for classification and regression tree algorithm, is used to form a tree (Figure 6.7).

Random Forest Model: RF is a well-known ML algorithm that is firmly rooted in the realm of supervised learning. It is useful in the field of ML for both classification and regression applications. This approach is based on the notion of ensemble learning, which is a strategy that includes combining numerous classifiers to solve complex issues and improve the model's performance. An "RF" classifier, as the name implies, is a classifier composed of several DTs constructed on varied subsets of the supplied dataset, and it utilizes their aggregate predictive potential to improve dataset accuracy. Rather than depending exclusively on a single DT, the RF aggregates forecasts from each tree and then creates a final prediction based on a majority vote of these projections. The presence of a higher number of trees inside the forest adds to improved accuracy and protects against overfitting. Because the RF mixes numerous trees to forecast the dataset's class/output, some DTs may predict the proper output while others may not (Figure 6.8).

6.4.3 Support Vector Machine

The SVM is a supervised ML algorithm that may be used for classification as well as regression tasks. While SVM can handle regression issues,

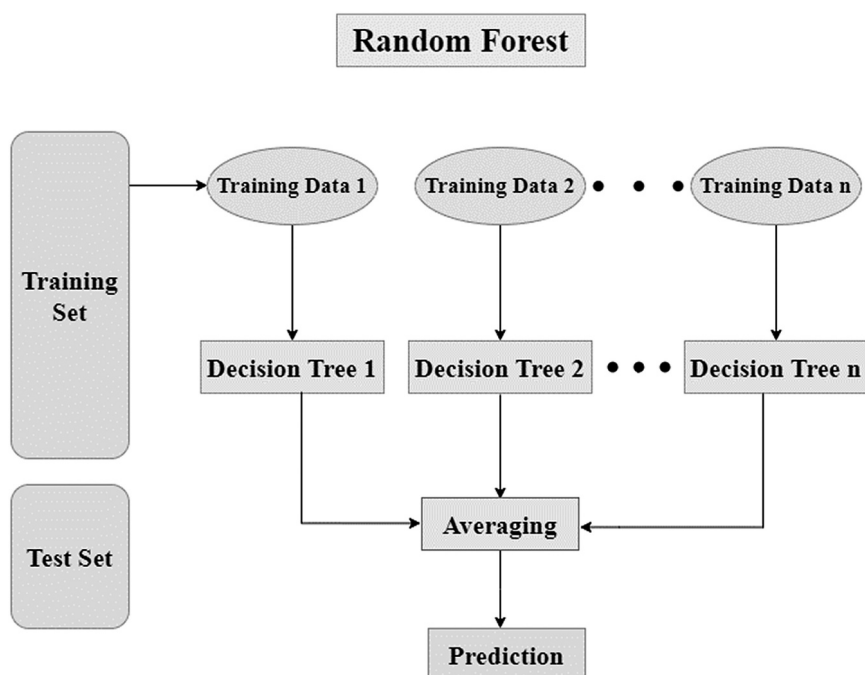
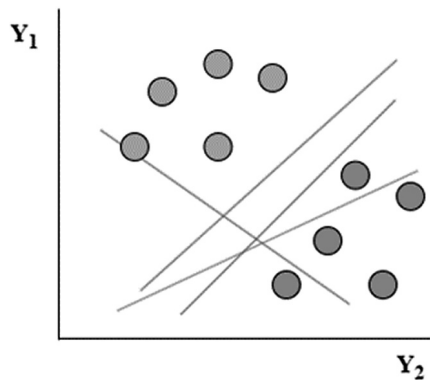


FIGURE 6.8
RF classifier.

it is best suited for classification jobs. Its major purpose is to find the best hyperplane inside an N-dimensional space capable of properly classifying data points in the feature space. The hyperplane's size is determined by the amount of input characteristics and is intended to maximize the gap between the nearest points of separate classes. When the number of features exceeds three, it becomes difficult to visualize the hyperplane. The SVM is a robust ML technique used for applications such as linear or nonlinear classification, regression, and even outlier identification. Text classification, picture classification, spam identification, handwriting recognition, gene expression analysis, face detection, and anomaly detection are just a few of the tasks that SVMs are used for. Because of their capacity to handle high-dimensional data and account for nonlinear interactions, their versatility and efficiency show through a wide range of applications. When attempting to discover the largest separating hyperplane among several classes within the target feature, the efficiency of SVM algorithms is clear. Let's consider two independent variables Y_1 and Y_2 , and one dependent variable, which is either a green circle or a yellow circle.

**FIGURE 6.9**

Linearly separable data points.

As seen in Figure 6.9, numerous lines (our hyperplane is a line in this case because we only have two input characteristics, Y_1 and Y_2) exist, thereby splitting our data points and conducting a classification between the red and blue circles. Mekni et al. uses an SVM algorithm as supervised learning for the prioritization of novel potential of the SARS-CoV-2 main protease inhibitor and explains how an SVM [39].

6.4.4 Neural Network Model

An NN is an AI technology that allows computers to interpret data like that of the human brain. It is classified under ML, especially a branch known as deep learning. This method organizes linked nodes or neurons into layered structures resembling the human brain. This design creates an adaptive system in which computers may learn from mistakes and gradually improve their performance. As a result, artificial NNs attempt to perform complicated tasks with greater precision, such as document summarization or facial recognition. There are various applications of NN, a few of them are listed next:

Natural Language Processing: Natural language processing (NLP) is the ability to analyze human-generated text in a manner that mimics human language comprehension. For computers to extract insights and comprehension from textual data and texts, NNs play a critical role. NLP has a variety of practical applications, such as the following:

- Giving automated virtual agents and chatbots more power.
- Organizing and categorizing textual stuff automatically.

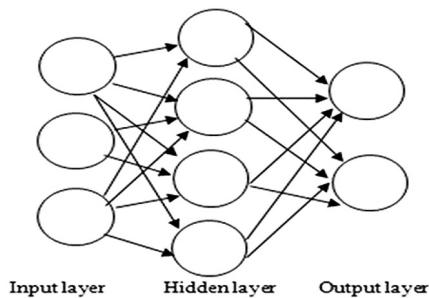
- Enabling business intelligence analysis of large documents like emails and forms.
- Indexing key terms that express attitude, such as good and negative social media comments.
- Creating document summaries and essays on certain subjects.

Computer Vision: Computer vision refers to a computer's ability to extract information and comprehend visual content from images and videos. Using NN, computers are now capable of distinguishing and classifying pictures in the same way that people do. Computer vision has practical applications in a variety of fields, including the following:

- Visual recognition in self-driving vehicles allows them to recognize road signs and other vehicles on the road
- Content moderation, which involves automating the identification and removal of potentially hazardous or improper content from picture and video collections
- Facial recognition for recognizing certain facial characteristics such as open eyes, spectacles, or facial hair
- Image labeling for features, such as company logos, clothes, safety equipment, and other image information

Convolutional Neural Networks (CNN): CNNs are a type of multi-layered NN that is specially designed for processing two-dimensional input, such as pictures and videos. CNNs are inspired by earlier work on time-delay neural networks (TDNN), which optimize learning calculations by weight sharing in a temporal dimension and are particularly suited for speech and time-series data processing applications [40]. CNNs were a significant milestone in the field of deep learning by effectively robustly training deep hierarchies of layers. These networks use an architectural approach that capitalizes on spatial and temporal correlations, minimizing the number of parameters that need to be learned and resulting in better performance than traditional feed-forward back-propagation training [41].

CNNs are distinguished by their low data preprocessing needs. Images are broken into tiny parts in CNNs, which serve as inputs to the lowest layer of the hierarchical structure. Information passes via several network levels, with each layer using digital filtering techniques to extract important data properties. This method gives the network some resistance to shifts, scaling, and rotations, owing to the local receptive field, which gives neurons or processing units access to basic properties, such as aligned edges and corners.

**FIGURE 6.10**

An ANN model.

Artificial Neural Network (ANN): An ANN is a network of linked nodes that is remote, like the huge network of neurons in the brain, as seen in Figure 6.10. Each circular node represents an artificial neuron [42], and each arrow symbolizes a link from one neuron's output to the input of another, which should (hopefully) be capable of handling this. An ANN is made up of three types of layers: input, hidden, and output. The hidden layer connects the input and output layers.

6.4.5 Ensemble Models

Ensemble learning is a comprehensive meta-strategy in ML that aims to improve predicted accuracy by combining predictions from many models. While there appears to be an infinite number of ensembles that may be built for predictive modeling, three approaches dominate the field of ensemble learning. In fact, these procedures have grown into entire disciplines of study, giving rise to a plethora of specialized techniques within each. Bagging, stacking, and boosting are the three primary classes of ensemble learning methods, and it is critical to understand each approach and include it in the predictive modeling project.

6.5 CASE STUDIES

6.5.1 Prediction of Crop Yield for a Specific Crop Type

Shahram Rezapour [18] and the rest of his team completed a study that focused on using ML approaches to predict agricultural productivity in arid and semi-arid regions. Their major goal was to address food security

challenges, particularly related to climate change's impact on food production. The case study in the issue is about estimating chickpea output in Kermanshah Province [43], which is located in Iran. The researchers used remote sensing data and meteorological information from synoptic stations to build prediction models as important inputs in their technique. They used ML techniques such as support vector regression (SVR) [44], RF [45], and extreme randomized trees (ERT) to analyze the data and provide exact estimates of rainfed agricultural productivity.

The research findings demonstrated that, when evaluating the optimum combination (OC) scenario, the RF strategy produced the lowest errors among the numerous approaches studied. Furthermore, in this situation, the RF technique indicated surface moisture, minimum air humidity, and subsurface humidity as the most important parameters for accurate forecasts in March, January, and January, respectively. Furthermore, the researchers suggested that future studies look at different crop kinds, prediction systems, and geographic areas other than chickpeas, broadening the scope to include other agricultural goods. In conclusion, Rezapour and his colleagues' work demonstrates the effectiveness of ML algorithms for predicting rainfed agricultural outputs. Their study provides useful insights into predicting chickpea output in Kermanshah Province, Iran, while emphasizing the need for accurate data collection and analysis in making well-informed agricultural decisions [46]. These findings contribute significantly to ongoing programs targeted at ensuring food security in arid and semi-arid countries.

6.5.2 Dealing with Extreme Weather Events

[47] his study is about assessing and forecasting drought conditions in Jaisalmer, India, using data-driven methodologies. It uses ML techniques, notably the random subspace (RSS) approach, to calculate the standardized precipitation index (SPI) over different periods. The data illustrate the RSS-M5P model's usefulness in capturing monthly SPI trends and providing accurate SPI predictions at several scales, including 3, 6, and 9 months.

The findings of this study have substantial practical implications for a variety of industries, including agriculture, hydrology, and meteorology. Drought prediction models produced by researchers are useful for activities such as optimizing irrigation schedules, modeling crop behavior, managing water resources, running reservoirs, and improving weather forecasts. It emphasizes the importance of developing comprehensive drought

management policies. To summarize, this case study focuses on the use of data-driven models to analyze and predict drought in Jaisalmer, India [48]. It calculates the SPI across multiple temporal scales using ML and the RSS approach. These models have practical implications in a variety of industries and can aid in the development of climate-smart farming practices.

6.6 DISCUSSION

The use of ML algorithms in crop production prediction has far-reaching consequences for the agricultural environment. We unleash the potential for more exact and timely production projections by harnessing these advanced computational methods, providing farmers and stakeholders with essential information about the performance of their crops. This results in better decision-making, resource optimization, and risk reduction. Furthermore, improved forecasting ability owing to weather, pests, or illnesses helps to increase food security and economic stability [49]. As ML models improve their forecast accuracy, the agricultural industry will become more resilient and sustainable, influencing global food production and assuring a more secure food supply for our rising population.

Several obstacles and limits must be considered while traversing the area of ML algorithms for agricultural yield projections. One of the most significant issues is data quality and availability [50], as successful projections rely largely on complete datasets that include parameters like weather, soil conditions, and historical yields. Furthermore, model interpretability remains a challenge, prompting research efforts to make these algorithms more visible and accessible for farmers and stakeholders. Another issue to consider is scalability, which is especially important in resource-constrained environments. To address these issues, future research should focus on multi-source data integration, real-time monitoring systems that can adapt to changing conditions, and ML solutions that are tailored for low-resource contexts. ML in crop-yield projections has practical applications in precision agriculture, where farmers may optimize resource allocation, insurance and risk assessment, financial planning, and supply chain management, improving logistics and minimizing waste. As this sector evolves, it has the potential to revolutionize agricultural practices and greatly contribute to global food security. Future research should focus on the interface of ML and climate change adaptation, allowing agriculture to adjust to changing weather patterns and environmental circumstances. More interpretable AI

models will boost confidence and usefulness, while ongoing developments in data-gathering methods, sensor technology, and precision agricultural tools will be critical. Finally, these activities will define agriculture's future by promoting sustainable practices and maintaining food security in an ever-changing world.

6.7 CONCLUSION

The agricultural sector must adopt new technology in order to satisfy the demands of a rising global population. Furthermore, timely advice for crop output forecasts is critical for farmers in developing successful production plans. ML frameworks provide significant insights by processing large datasets and analyzing the resulting data. These tools make it easier to create models that represent the interactions between numerous elements and activities. Furthermore, they allow for the prediction of future events in certain contexts. The current study focuses on a wide range of criteria used in the selected studies, with a particular emphasis on data availability and research scope. The majority of the cited works investigate agricultural production predictions using ML algorithms; however, they differ in terms of the vast variety of features used. These investigations also revealed differences in crop kinds, geographic areas, and intensity levels. The characteristics used are determined by the dataset's accessibility and the study aims. While it is difficult to determine the ultimate best technique based on present data, the widespread use of some ML algorithms and their promising performance is remarkable. Specifically, traditional ML architectures like linear regression, RF, and NN show potential. Some deep learning models, such as deep neural networks (DNN), CNN, and long short-term memory (LSTM), are used in crop-yield estimates in addition to these techniques. Future studies should explore feature selection algorithms in combination with existing high-performing models to get a clear conclusion about the best-performing model.

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